Dual Constraints Safe Reinforcement Learning for Autonomous Driving

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Abstract

We study safe reinforcement learning for urban autonomous driving in the MetaDrive "European Town" scenario. Within a unified PPO framework, we compare four agents: reward-only PPO [3], a Lagrangian variant PPO-Lag [2], Long- and Short-Term Constraints RL (LSTC) [1], and our PID-enhanced LSTC-PID. LSTC-PID keeps the dual long- and short-term safety constraints of LSTC but replaces the hand-tuned integral multiplier update with a PID controller and a trajectory-level correction step, which together adapt the strength of safety penalties more quickly and stably in high-variance driving scenes. On MetaDrive European Town, both LSTC and LSTC-PID consistently reduce collision, off-road, and traffic-rule violation rates compared to PPO and PPO-Lag while achieving comparable episodic return. Across three representative maps, LSTC-PID provides the best safety–return trade-off, yielding the lowest safety costs with only marginal loss in task performance.

1 Methodologies

We study how increasingly safety-aware variants of PPO behave in a dense urban driving benchmark. On the MetaDrive "European Town" scenario, we compare: (i) reward-only PPO, (ii) a long-term Lagrangian variant (PPO-Lag), (iii) a trajectory-level correction layer (LSTC), and (iv) an enhanced version with PID multipliers (LSTC-PID). We now formalize the task as a CMDP, describe the algorithms, and give the reward, costs, and training configuration.

Constraint Markov Decision Process (CMDP)

We model town driving as a CMDP $(\mathcal{S}, \mathcal{A}, P, r, \{c_i\}_{i=1}^m, \gamma)$ with continuous actions $a_t \in [-1, 1]^2$ (steering and longitudinal control) and discount $\gamma = 0.99$. An episode has horizon $H = 1000$ and generates a trajectory $\tau = (s_0, a_0, \ldots, s_{H-1}, a_{H-1}, s_H)$ under policy $\pi_\theta(a \mid s)$, where $s_t \in \mathbb{R}^{49}$ is the MetaDrive observation (lidar plus ego features). The expected return and discounted costs are

$$J(\theta) = \mathbb{E}_\theta\Big[\sum_{t=0}^{H-1} \gamma^t r(s_t, a_t)\Big], \qquad J_i^{\text{cost}}(\theta) = \mathbb{E}_\theta\Big[\sum_{t=0}^{H-1} \gamma^t c_i(s_t, a_t)\Big]. \tag{1}$$

We seek to maximize $J(\theta)$ subject to $J_i^{\text{cost}}(\theta) \le d_i$, using the standard Lagrangian relaxation

$$\mathcal{L}(\theta, \lambda) = J(\theta) - \sum_{i=1}^m \lambda_i \big(J_i^{\text{cost}}(\theta) - d_i\big). \tag{2}$$

PPO and PPO-Lag baselines

We next describe the policy optimization baselines used in all experiments.

PPO. We use standard PPO as the base algorithm. Let θ_{old} denote policy parameters before an update and

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \tag{3}$$

the importance ratio. Given reward advantages $\hat{A}_t$, the clipped surrogate is

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big], \tag{4}$$

with clip ratio ϵ. We add standard value and entropy terms (omitted here for brevity).

PPO-Lag. To handle long-term safety, PPO-Lag augments PPO with cost advantages. For each cost c_i we maintain a cost value function and compute cost advantages $\hat{A}_t^{\text{cost},i}$. The surrogate corresponding to the first-order Lagrangian (2) is

$$L^{\text{Lag}}(\theta, \lambda) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big) - \sum_{i=1}^{m} \lambda_i \hat{A}_t^{\text{cost},i}\Big]. \tag{5}$$

Multipliers are updated from empirical costs on a slower time scale:

$$\lambda_i \leftarrow \Big[\lambda_i + \eta_\lambda(\hat{J}_i^{\text{cost}} - d_i)\Big]_+, \quad i = 1, \ldots, m, \tag{6}$$

where $\hat{J}_i^{\text{cost}}$ is the discounted cost of constraint i in the current batch, η_λ is a step size, and $[\cdot]_+$ projects onto $\mathbb{R}_+$. Further details can be found in Ray et al. [2]. In summary, PPO-Lag uses the same policy architecture and optimization as PPO, but augments the surrogate with cost advantages and maintains one scalar multiplier λ_i per primitive cost signal c_i.

Trajectory-level correction

PPO-Lag penalizes expected long-term violations but does not explicitly edit risky actions. LSTC adds a trajectory optimizer with *two* safety constraints; our LSTC-PID variant keeps the same dual formulation but changes how the multipliers are updated.

Dual long- and short-term constraints. Whereas PPO-Lag maintains a separate multiplier for each primitive cost c_i, LSTC aggregates them and introduces two trajectory-level constraints. Inspired by Hu et al. [1], we introduce two trajectory-level costs. Let

$$c_{\text{long}}(s_t, a_t) = c_{\text{coll}}(s_t, a_t) + c_{\text{off}}(s_t, a_t) + c_{\text{rule}}(s_t, a_t). \tag{7}$$

The long-term cost is the discounted sum $C_{\text{long}}(\tau) = \sum_{t=0}^{H-1} \gamma^t c_{\text{long}}(s_t, a_t)$. To capture local bursts of risk, we define a short-term cost over a sliding window of size H_s:

$$C_{\text{short}}(\tau) = \sum_{t=0}^{H-1} \gamma^t \frac{1}{H_s} \sum_{k=0}^{H_s-1} c_{\text{long}}(s_{t+k}, a_{t+k}), \tag{8}$$

truncated near episode end. We write $C_j(\theta) = \mathbb{E}_\theta[C_j(\tau)]$ for $j \in \{\text{long}, \text{short}\}$ and optimize the dual objective

$$\mathcal{L}_{\text{LSTC}}(\theta, \lambda) = J(\theta) - \lambda_{\text{long}}\big(C_{\text{long}}(\theta) - d_{\text{long}}\big) - \lambda_{\text{short}}\big(C_{\text{short}}(\theta) - d_{\text{short}}\big). \tag{9}$$

Here $\lambda_{\text{long}}, \lambda_{\text{short}} \geq 0$ are multipliers for the long- and short-term constraints with budgets d_{long} and d_{short}. In practice, LSTC reuses the PPO-Lag surrogate but replaces the primitive costs J_i^{cost} by the aggregated long- and short-term costs C_{long} and C_{short}. Intuitively, C_{long} controls the overall number of failures, while C_{short} penalizes trajectories that enter locally unsafe regions.

Trajectory-level correction and PID multipliers. Given a sampled trajectory τ and current multipliers $(\lambda_{\text{long}}, \lambda_{\text{short}})$, we introduce corrected actions $\tilde{a}_t$ and minimize

$$\mathcal{J}(\tilde{\mathbf{a}}; \lambda) = -\sum_{t=0}^{H-1} \gamma^t r(s_t, \tilde{a}_t) + \lambda_{\text{long}} C_{\text{long}}(\tilde{\mathbf{a}}) + \lambda_{\text{short}} C_{\text{short}}(\tilde{\mathbf{a}}) + \beta \sum_{t=0}^{H-1} \|\tilde{a}_t - a_t\|_2^2, \tag{10}$$

where costs are computed along $(s_t, \tilde{a}_t)$ and $\beta > 0$ controls the trust region around the policy output. We perform a few gradient steps on $\tilde{\mathbf{a}}$ and treat the final $\tilde{a}_t$ as the executed actions; both during training and evaluation the environment receives these corrected actions, so the trajectory editor acts as a runtime safety filter on top of the PPO policy.

To adapt the dual variables more stably than the integral update (6), LSTC-PID replaces it with a discrete-time PID controller [4]. For $j \in \{\text{long}, \text{short}\}$ and iteration k, this is the method that we propose that replace the update role of λ.

$$\lambda_j^{(k+1)} = \left[K_P e_j^{(k)} + K_I I_j^{(k+1)} + K_D D_j^{(k)} \right]_+, \tag{11}$$

where $e_j^{(k)}$ is the deviation of the empirical cost C_j from its budget, $I_j^{(k+1)}$ accumulates past deviations, and $D_j^{(k)}$ is a finite-difference estimate of their change. The proportional, integral, and derivative gains (K_P, K_I, K_D) are fixed across runs and given in the appendix and accompanying code. Compared to LSTC, LSTC-PID thus keeps the same dual objective (9) and trajectory correction (10), and differs only in how the multipliers are updated.

Reward, costs, and training setup

We follow the reward design of Hu et al. [1] and denote the per-step reward by $r(s_t, a_t)$, decomposed as

$$R_t = c_1 R_{\text{dis},t} + c_2 R_{\text{speed},t} + c_3 R_{\text{yaw},t} + c_4 R_{\text{steer},t} + R_{\text{term},t}, \tag{12}$$

with $(c_1, c_2, c_3, c_4) = (1.0, 0.1, -0.4, -0.4)$. Here $R_{\text{dis},t}$ rewards longitudinal progress along the route, $R_{\text{speed},t}$ encourages driving near a target speed ($v_t \approx v_{\max}$), and $R_{\text{yaw},t}$ and $R_{\text{steer},t}$ penalize rapid changes in heading and steering, promoting smooth lane-keeping. $R_{\text{term},t}$ is a sparse terminal term that gives a positive bonus when the destination is reached and a negative penalty when the vehicle leaves the lane or the episode terminates due to failure.

Safety costs are binary indicators from MetaDrive events:

$$c_{\text{coll}}(s_t, a_t) = \mathbb{I}\{\text{collision at } t\}, \tag{13}$$

$$c_{\text{off}}(s_t, a_t) = \mathbb{I}\{\text{out-of-road at } t\}, \tag{14}$$

$$c_{\text{rule}}(s_t, a_t) = \mathbb{I}\{\text{traffic-rule violation at } t\}. \tag{15}$$

Budgets d_i are chosen so that the target expected number of violations per episode is at most one, and are shared across PPO-Lag, LSTC, and LSTC-PID.

All agents share standard PPO hyperparameters (discount $\gamma = 0.99$, GAE parameter $\lambda_{\text{GAE}} = 0.95$, clip ratio $\epsilon = 0.2$, entropy coefficient 0.01) and are trained for 10^7 environment steps. Full hyperparameter lists and implementation details, including hardware and configuration files for each method, are provided in the appendix and our open-source repository at `https://github.com/WilliamQiuzy/SafeMeta-RL`.

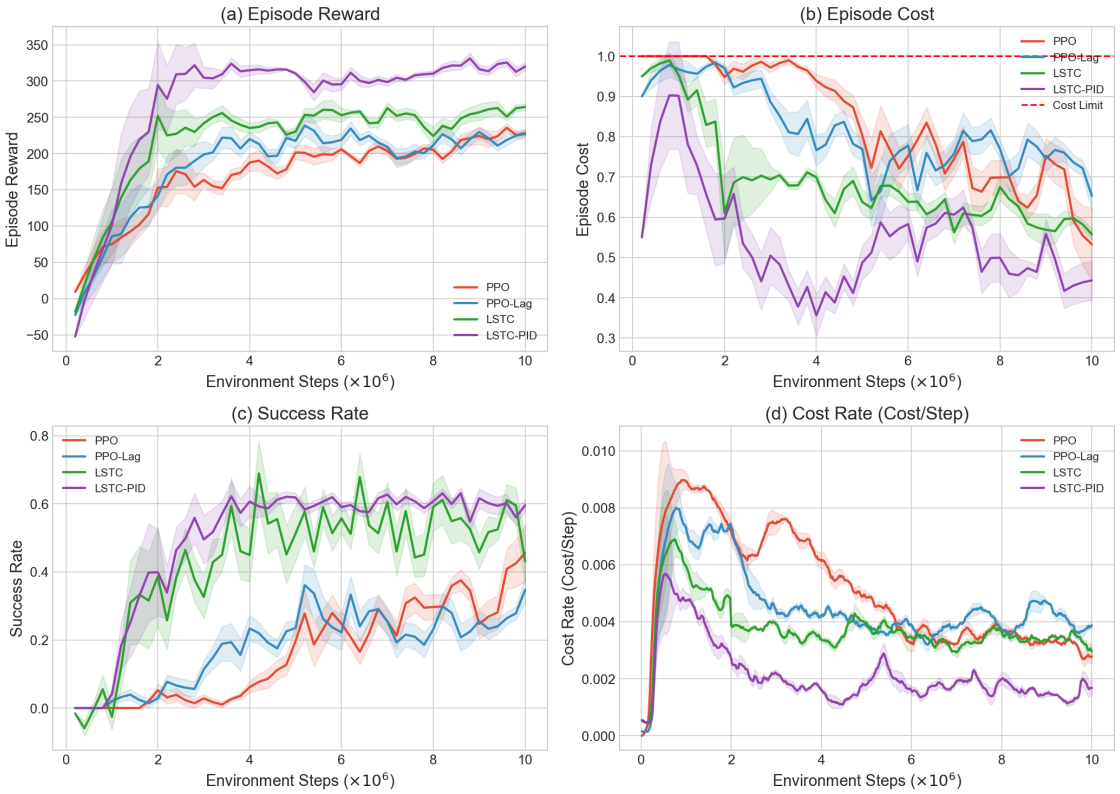


Figure 1: Training curves showing episode reward and safety metrics for PPO, PPO-Lag, LSTC, and LSTC-PID on the European Town benchmark.

2 Results

We evaluate PPO, PPO-Lag, LSTC, and LSTC-PID in the MetaDrive European Town map using both performance and safety metrics.

2.1 Evaluation metrics

For each method we report the average episodic return $\mathbb{E}[\sum_t \gamma^t r(s_t, a_t)]$ and success rate, defined as the fraction of episodes that reach the destination without catastrophic failure. We also track per-episode safety costs derived from c_{coll}, c_{off}, and c_{rule}, as well as their empirical violation rates. All scalars are averaged over multiple evaluation episodes on held-out maps; when running several random seeds we aggregate statistics across seeds and report the mean.

2.2 Reward curves

To study learning dynamics, Figure 1 shows the average episode return and safety metrics over environment steps for all four algorithms, smoothed with a moving window.

We observe that LSTC and LSTC-PID reach similar or slightly lower asymptotic return than PPO, while PPO-Lag tends to under-explore and plateaus at a lower return level. At the same time, LSTC and LSTC-PID converge to substantially lower cost levels than both PPO and PPO-Lag.

We also note that the training curves remain somewhat noisy, reflecting the high-variance traffic scenarios and our finite training budget.

Table 1: Success rate (↑) and episode cost (↓) on three representative European Town scenarios.

Method	Success rate ↑			Episode cost ↓		
	InRamp	Slet	Rodab	InRamp	Slet	Rodab
PPO	98%	78%	36%	3.1	3.7	4.7
PPO-Lag	98%	100%	55%	0.4	1.82	1.82
LSTC	100%	100%	73%	0.2	0.56	0.88
LSTC-PID (ours)	100%	100%	78%	0.5	0.5	0.70

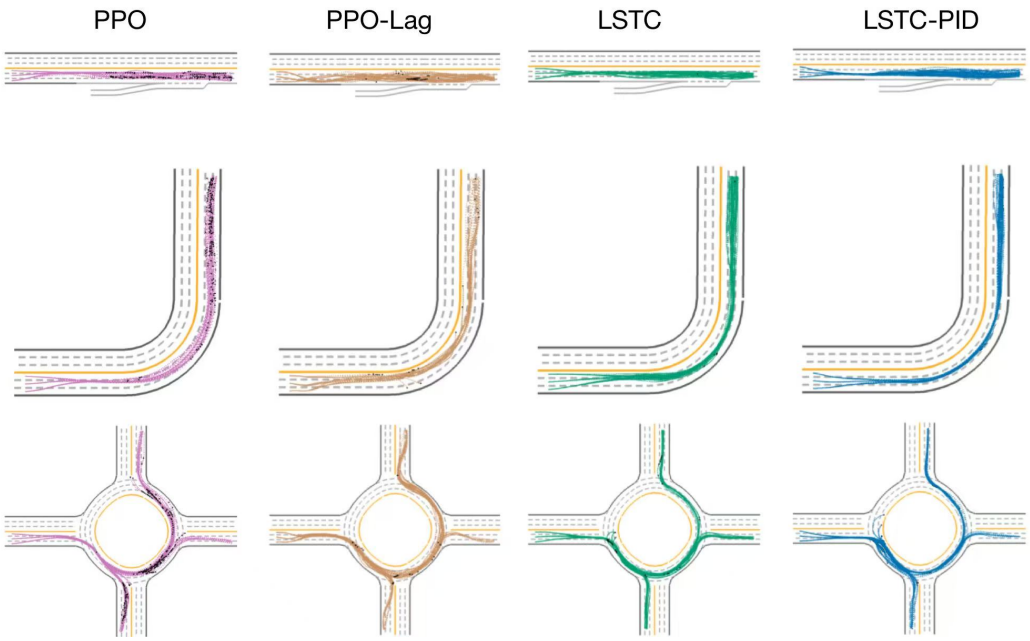


Figure 2: Representative trajectories in the European Town map comparing PPO, PPO-Lag, LSTC, and LSTC-PID on a multi-vehicle intersection scenario.

2.3 Quantitative comparison

Table 1 summarizes the main quantitative results in the style of the original LSTC paper: success rates and episode costs for several European Town scenarios. Across all three maps, LSTC and LSTC-PID substantially reduce safety cost compared to PPO and PPO-Lag, with LSTC-PID achieving the best overall safety–return trade-off: collision and off-road costs are consistently lowest, while success rates remain close to the PPO baseline.

2.4 Trajectory examples

Figure 2 visualizes representative trajectories for all four methods on a dense-intersection scenario. PPO often approaches intersections aggressively and performs late lane changes, whereas LSTC and especially LSTC-PID exhibit smoother braking profiles and maintain larger safety margins around other vehicles.

References

[1] X. Hu, P. Chen, Y. Wen, B. Tang, and L. Chen, "Long and Short-Term Constraints Driven Safe Reinforcement Learning for Autonomous Driving," *arXiv preprint* arXiv:2403.18209, 2024.

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